D-affinity

sheaf global sections, NOT D-module pish

forget D_X-action. $\mathcal{O}_X$-action...

Def^n X is D-affine if:

- $\Gamma(X,-): DMod(X) \longrightarrow Vect$ is exact
- Γ is conservative ①. i.e. $\Gamma(M)=0$ iff $M=0$

Spoiler: $\mathbb{P}^1$ is D-affine. $\Rightarrow$ $\mathcal{O}(n)$ can be - D-module iff $n=0$.

② conservative if $\Gamma(\phi)$ iso then ϕ is iso

assume exactness,

$\phi: M \to N \to \mathrm{coker}(\phi) \to 0$

$\Gamma(M) \xrightarrow{\Gamma(\phi)} \Gamma(N) \to \Gamma(\mathrm{coker}\,\phi) \to 0$ is

$M \xrightarrow{\phi} M$ then $\Gamma(M) \to \Gamma(M)$

if $\Gamma(M)=0$, then $\Gamma(\phi)$ is iso.

Gives ②, ϕ iso $\Rightarrow M=0 \Rightarrow$ ①

Remark: similar kinds arguments

Γ conservative $\Longleftrightarrow$ D-modules are globally generated

i.e. $D_X \otimes_{\Gamma(D_X)} \Gamma(M) \to M$ is surjective $\forall M$

(Γ right exact)

Examples: ① $\mathbb{P}^1$, $\mathbb{P}^n$, BB. G/B

② X affine

③ G/H projective homogeneous spaces

Conjectures: this is a complete list

Prop: If X is D-affine, then $\Gamma(X,-): DMod(X) \longrightarrow Mod(D(X))$ is an equivalence of categories.

left modules for $\Gamma(X, D_X)$

Def^n adjoint functors $\mathcal{C} \underset{G}{\overset{F}{\rightleftarrows}} \mathcal{D}$ $\quad T = GF \in End(\mathcal{C})$ monad

$T \circlearrowleft$

"mult": $\mu: T^2 \xrightarrow{\mu} T$ $\qquad GFGF \to GF$

"unit": $\eta: id_{\mathcal{C}} \longrightarrow T = GF$ unit of adjunction

from the adjunction

coinit applied here.

Module for T: object $c \in \mathcal{C}$ + "action map" $T(c) \xrightarrow{\alpha} c$

satisfy usual identities: $c \xrightarrow{\eta} T(c) \xrightarrow{\alpha} c$, $\alpha \circ \eta = id_c$ ...

Thm (Barr-Beck) In this setup: assume
- $\mathcal{D}$ has coequalizers
- G is conservative, preserves coeq. (exactness)

There is an equivalence:

$$\begin{tikzcd} \mathcal{C} \arrow[r, shift left, "F"] & \mathcal{D} \arrow[l, shift left, "G"] \arrow[dl, "\tilde{G}"] \\ \mathrm{Mod}_{\mathcal{C}}(T) \arrow[u, "\simeq"] & \end{tikzcd}$$

forget T-mod str.

$\tilde{G}(d) = G(d)$ has a natural T-mod str.

abstract nonsense.

① Proof prop: In abelian cat, exactness $\Rightarrow$ preserves coequalizers.

$$\mathrm{coeq}\left(X \underset{g}{\overset{f}{\rightrightarrows}} Y\right) = \mathrm{coker}\left(X \xrightarrow{f-g} Y\right)$$

So, $\Gamma = G$ is conservative, pres. coeq, and $\mathcal{D}$ is abelian (has coeq)

Check: $T = GF = \Gamma(p(-))$
$= \Gamma(D_X$

$(F,G) = (p^*, p_*)$ $\quad p: X \to \alpha$

$V \mapsto D_X \otimes V$

Do as exercise

Claim: $T^2 \longrightarrow T$

$$\begin{array}{ccc} T^2 & \longrightarrow & T \\ \| & & \| \\ -\otimes D(X)\otimes D(X) & \xrightarrow[\text{mult.}]{} & -\otimes D(X) \end{array}$$

$\mathrm{Hom}(D_X \otimes k, M) = \mathrm{Hom}(k, \Gamma(M))$

$\mathrm{Hom}_{D_X}(D_X, M) = \Gamma(M)$

comes from counit.

$FG \longrightarrow id$

act using $D(X)$ action on $\Gamma(M)$

$D_X \otimes_{D(X)} \Gamma(M) \longrightarrow M$

Input $M = D_X$

$D(X) \otimes D(X) \longrightarrow D(X)$

act $\downarrow$ mult.

1-cat. presentation

derived / ∞-cat: resolution.

Ex: Idea behind Barr-Beck: monad T $\rightsquigarrow$ build monadic resolutions of T-modules.

F: free functor, $\mathrm{Set} \to \mathrm{Mon}$

G: forgetful: $\mathrm{Mon} \to \mathrm{Set}$

$T = GF$ "set of words"

- $\mathrm{colim}\left(T^2(c) \underset{\alpha}{\overset{\mu}{\rightrightarrows}} T(c)\right) \xrightarrow{\sim} c$

- $\mathrm{colim}\, T^2(Gd) \rightrightarrows T(Gd) \xrightarrow{\varepsilon} Gd$ also presentation

conditions: G cons. right ⊗.

$\boxed{c \text{ is } \underline{\text{presented}} \text{ by monadic presentation}}$

$\eta : \mathrm{id} \to T$ "single letter words"

$\mu : T^2 \to T$ interp words of words as word $(xy)(yz) = xyyz$

~~Ch~~ module: set S + $T(S) \to S$ way to turn words into letters
i.e. "multiplication"

Check: Say T-module S

word of word $\mapsto$ word.

$$\mathrm{colim}\left(T^2(S) \rightrightarrows T(S) \right) \to S$$

$(xy)(yz) \mapsto xyyz$
$\qquad\qquad\qquad\quad \| $
$\qquad\qquad\qquad x \cdot y \;\; y \cdot z$

word of word $\mapsto$ resolve inner word $(x\,y\,z) \mapsto xyz$
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \| $
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad x \cdot y \cdot z$

Resolution is big, but tautological / canonical.

Enough: comonadic Barr-Beck. $p : \mathcal{U} \to X$ open cover of topl space X
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \| $
$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \coprod \mathcal{U}_i$

(p^*, p_*) adjunction

comonadic resolution is Čech complex for this cover.

$$\mathrm{colim}\left(\cdots\cdots T^3(c) \rightrightarrows T^2(c) \rightrightarrows T(c) \right) \to c \qquad T =$$

$$\cdots \rightrightarrows D_X \otimes_{D_X} \Gamma(D_X \otimes_{D_X} \Gamma(M)) \rightrightarrows D_X \otimes_{D(X)} \Gamma(M) \to M$$

D-affinity.

"Pf" Barr-Beck:

$$\begin{tikzcd} \mathcal{C} \arrow[r, shift left, "F"] & \mathcal{D} \arrow[l, shift left, "G"] \arrow[dl, "\tilde{G}"] \\ \mathrm{Mod}_{\mathcal{C}}(T) \arrow[u] & \end{tikzcd}$$

$\tilde{G}(d) = G(d)$ $\qquad$ $G\,\mathrm{id}_{\mathcal{D}}\,d$

$\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad